

MA546 Introduction to Quantitative Finance

Assignment 4

1. Let $\boldsymbol{\mu} \in \mathbb{R}^n$, $\boldsymbol{\Sigma} \succ 0$, $A = \mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \mathbf{1}$, $B = \mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$, $C = \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$, $\Delta = AC - B^2 > 0$.
 - (a) Derive the GMV weights $\mathbf{w}_{\text{GMV}}$, mean μ_{GMV} , and variance σ_{GMV}^2 using KKT conditions.
 - (b) Verify $\sigma_{\text{GMV}}^2 = 1/A$ directly.
 - (c) Using the data below compute $\mathbf{w}_{\text{GMV}}$, μ_{GMV} , σ_{GMV} numerically.

$$\boldsymbol{\mu} = \begin{pmatrix} 6\% \\ 10\% \\ 14\% \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} 0.010 & 0.002 & 0.001 \\ 0.002 & 0.0225 & 0.003 \\ 0.001 & 0.003 & 0.040 \end{pmatrix}.$$

2. For the same three-asset data, return target m :
 - (a) Derive the KKT conditions for the budget-and-return constrained Markowitz QP and show $\mathbf{w}^*(m) = m \mathbf{g} + \mathbf{h}$ for vectors $\mathbf{g}, \mathbf{h}$ you must identify.
 - (b) State and prove the Two-Fund Theorem.
 - (c) Compute $\mathbf{w}^*(10\%)$ numerically using A, B, C, Δ from Q1. Also give $\sigma(10\%)$.
3. Add long-only bounds $w_i \geq 0$ to the Markowitz QP.
 - (a) Write the full KKT conditions (budget, return, non-negativity). Define each multiplier and state its sign.
 - (b) Explain complementary slackness for the bound multipliers. What does $\nu_i > 0$ mean geometrically?
 - (c) For the two-asset case: $\mu_1 = 6\%$, $\mu_2 = 12\%$, $\sigma_1 = 10\%$, $\sigma_2 = 18\%$, $\rho = 0.3$, $m = 8\%$. Find $\mathbf{w}^*$ and verify the KKT conditions.
4. Select at most $K = 2$ assets from $\{A_1, A_2, A_3\}$ to minimise variance, subject to $\mathbf{1}^\top \mathbf{w} = 1$, $\boldsymbol{\mu}^\top \mathbf{w} \geq 9\%$, $w_i \geq 0$. Same three-asset data as Q1.
 - (a) Write the cardinality-constrained MIQP using binary variables $z_i \in \{0, 1\}$ and explain the big-M constraint $w_i \leq z_i$.
 - (b) Enumerate all $\binom{3}{2} = 3$ two-asset subproblems and identify the optimal pair.

5. Portfolio $(w_1, w_2) = (0.6, 0.4)$, target $\tau = 1\%$, six scenarios:

t	1	2	3	4	5	6
$r_{1t} (\%)$	3	1	-2	4	-3	2
$r_{2t} (\%)$	5	-1	2	-3	1	4

- (a) Compute $R_t = 0.6r_{1t} + 0.4r_{2t}$ for each t , then calculate $\widehat{\text{SV}}(w; \tau)$.
 - (b) Write the semivariance-minimisation QP (general n, T). Show it is convex.
 - (c) Prove $\text{SV}(\mathbf{w}; \tau) \leq \text{Var}(R_P)$ when $\tau = \mu_P$.
6. A portfolio has $\mu_P = 0.05\%/ \text{day}$, $\sigma_P = 2\%/ \text{day}$, value Rs. 10,00,000.
 - (a) Give the formal definition of $\text{VaR}_\alpha(L)$ as a quantile. State its CDF geometric interpretation.
 - (b) Ten historical losses (in %) sorted ascending: -2.0, -1.3, -0.2, 0.5, 0.8, 1.1, 1.8, 2.1, 3.5, 4.2. Compute $\widehat{\text{VaR}}_{90\%}$.
7. Ten sorted historical losses (ascending): -2.0, -1.3, -0.2, 0.5, 0.8, 1.1, 1.8, 2.1, 3.5, 4.2 (in %).
 - (a) Define $\text{CVaR}_\alpha(L)$ and state the integral formula.
 - (b) Compute $\text{CVaR}_{90\%}$ and $\text{CVaR}_{80\%}$ from the scenarios. Verify $\text{CVaR}_\alpha > \text{VaR}_\alpha$ in both cases (use VaR values from Q9d).
 - (c) For $\mu_P = 0$, $\sigma_P = 2\%$, compute $\text{CVaR}_{99\%}$ under normality (use $\phi(2.326) = 0.02667$).
 - (d) Prove $\text{CVaR}_\alpha \geq \text{VaR}_\alpha$ in general.